

RTI Admin Console: A Step-by-Step Guide for Connex Micro Applications

This guide provides an overview of how to use the RTI Admin Console to monitor topics, subscribe to data, and publish data within a remote Connex Micro application.

Note: This guide explains how to validate an example application using a simple DDS-XML. To use a production DDS-XML, refer to the section in the [annex](#) on configuring Admin Console QoS.

Prerequisites

RTI Admin Console is part of the tools included with RTI Connex Drive. Please make sure this is installed before following this guide.

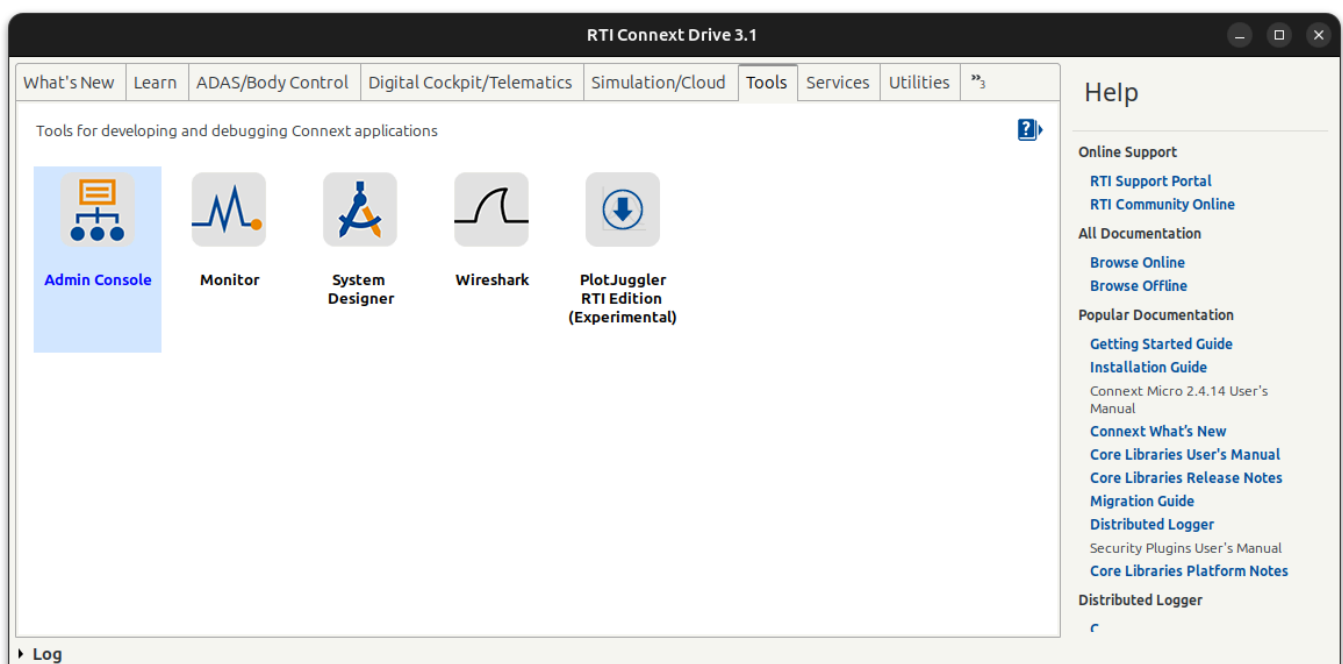
Also, the Graphical Data Publishing feature uses Connex Python API to populate the data. Make sure there is a Python interpreter installed. Do not use the Windows Store interpreter. Check this [Knowledge Base Article](#) in case you run into problems regarding Python.

1. Discovering a Connex Micro Application

1.1 Launching Admin Console

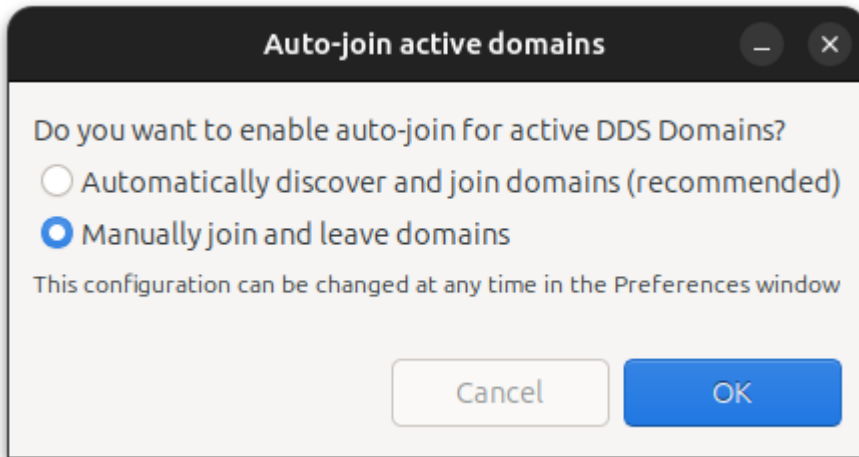
Execute the RTI Launcher from your Connex installation.

You can find Admin Console within the “Tools” tab.



The first time you run Admin Console, the following prompt will show. Choose the option “Manually join and leave domains”.

If this message doesn't appear, that's OK. We'll apply the right configuration in the following sections.



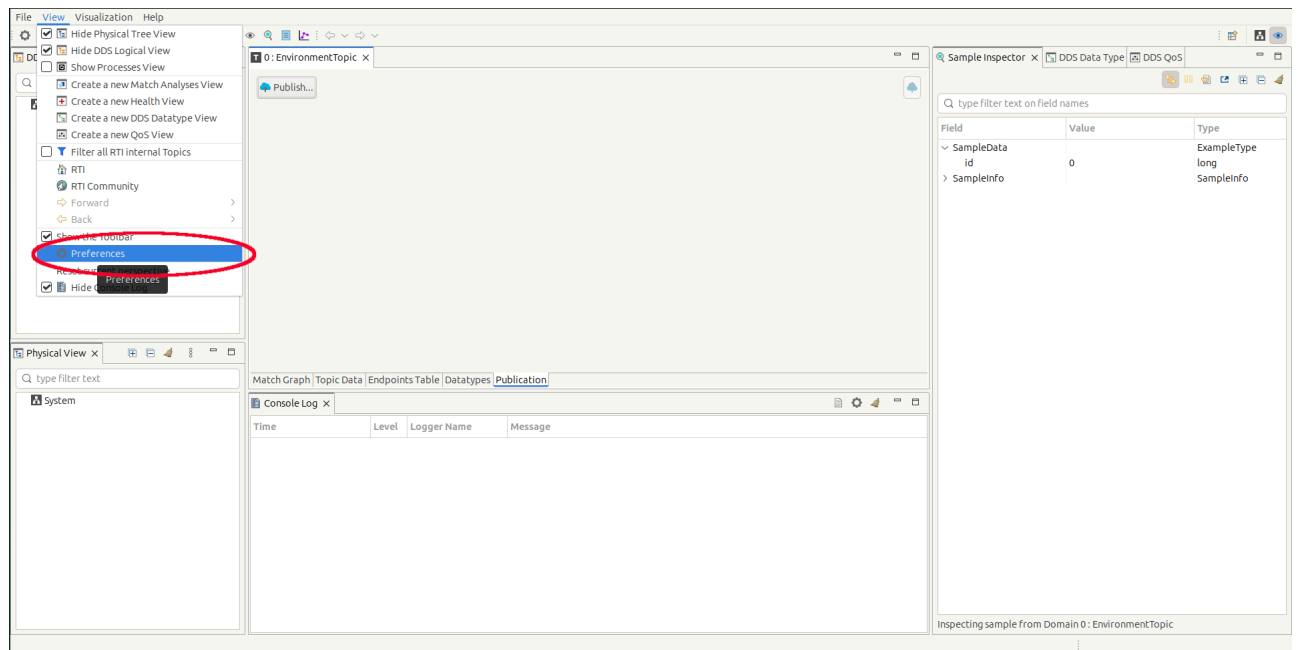
1.2 Discovering Your Application

The following steps explain how to configure Admin Console to discover the Connex Micro application.

Note: if the Micro application uses multicast, only the Domain configuration in the below is needed.

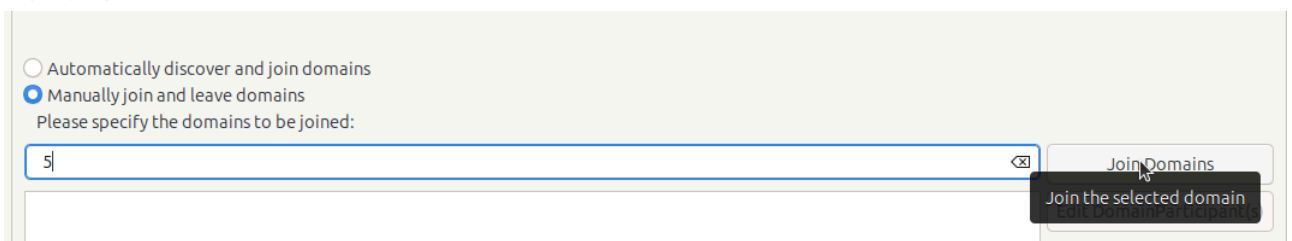
1.2.1 Configure custom Initial Peers

1. Go to View > Preferences.

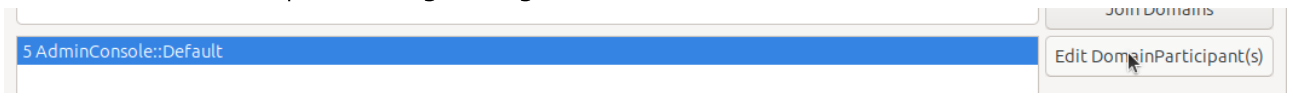


2. In the Preferences dialog, select the Administration tab.

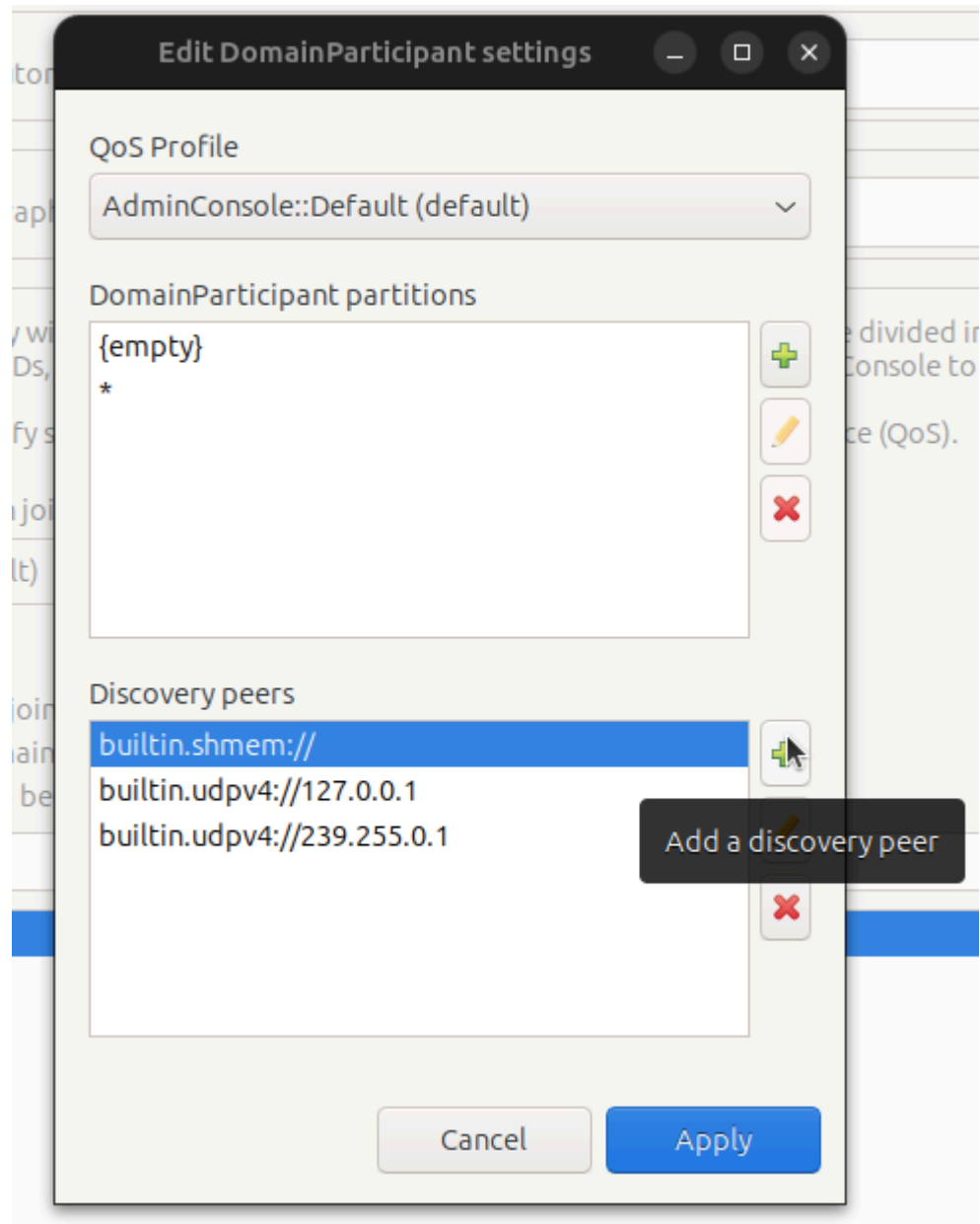
3. Select "Manually join and leave domains". Specify the domain of your application. Then click "Join Domains"



4. **(Only if you are not using multicast)**, click the domains and then "Edit DomainParticipant(s)" to open the Edit DomainParticipant settings dialog.

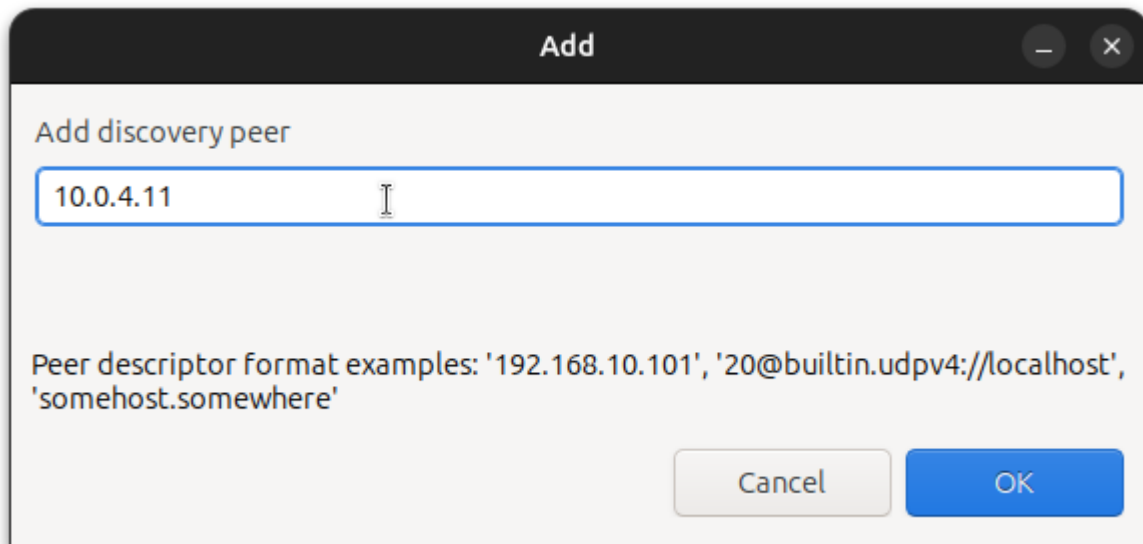


5. **(Only if you are not using multicast)** Add the IP address of the Micro application that you want to



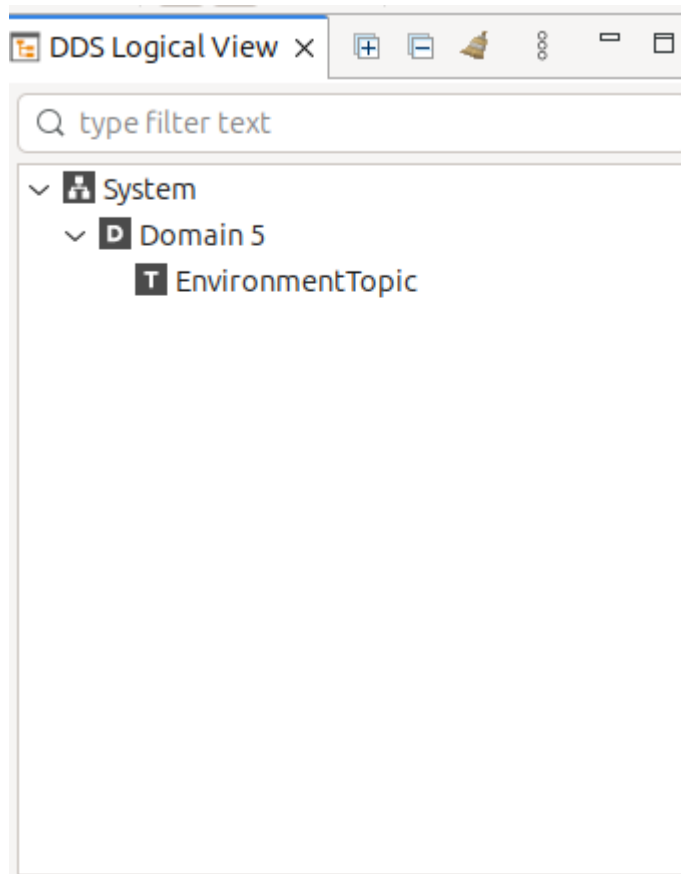
connect in this dialog.

After adding the peer address:

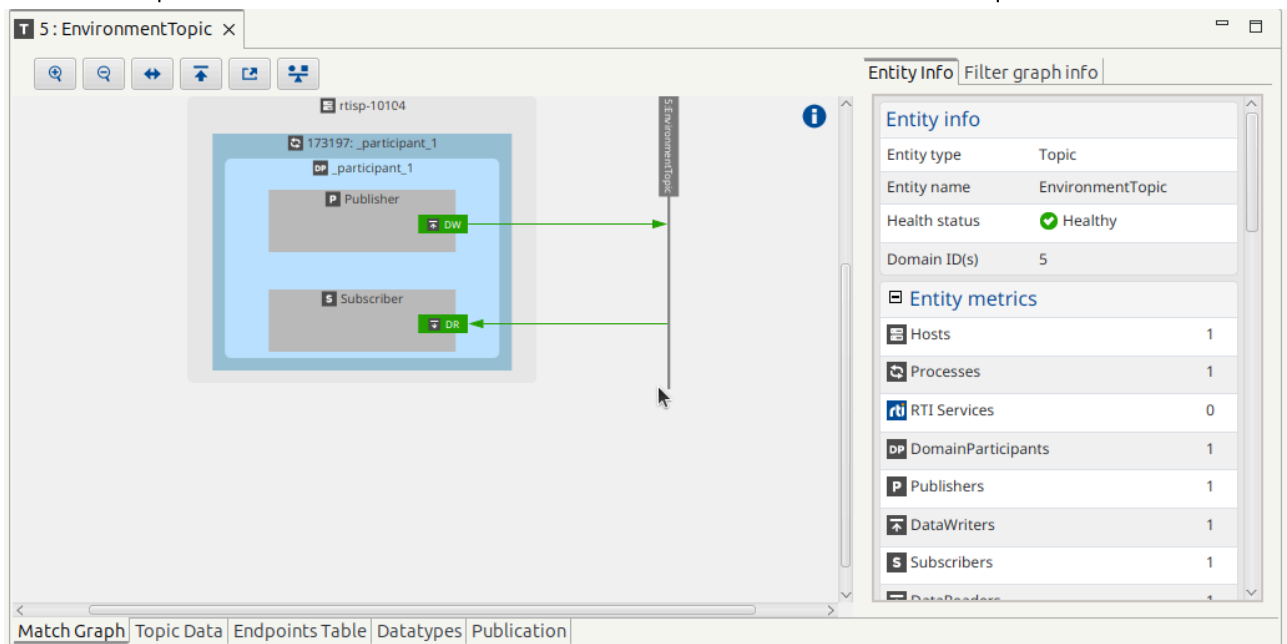


1.3 Navigating the DDS Logical View

1. Once a domain is joined, the "DDS Logical View" will display a list of discovered Topics
2. Expand the "Domain" nodes to see all active topics in the domain.



- Click on a topic to see the associated DataWriters and DataReaders in the main pane.

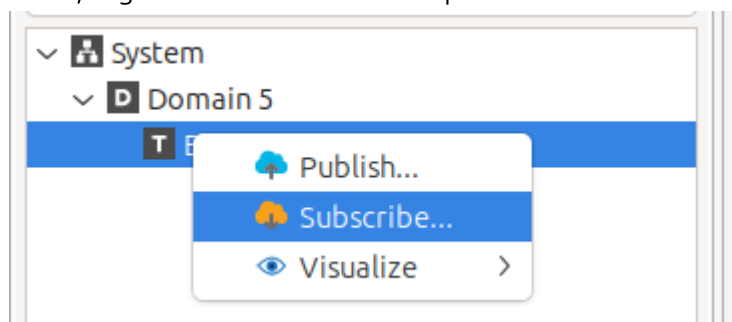


2. Subscribing to Data Using Admin Console

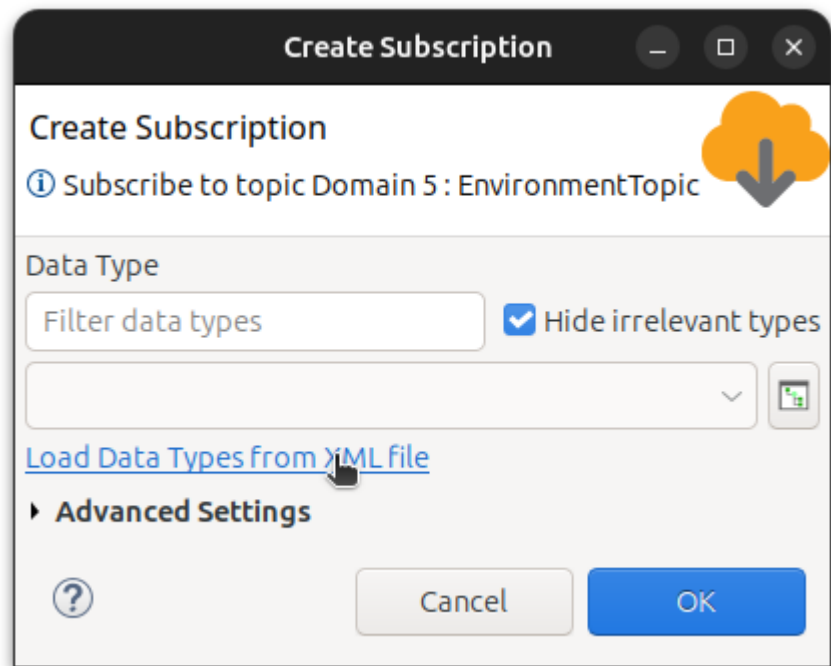
Admin Console enables you to dynamically subscribe to topics and view the data being published by your Connex Micro application.

2.1 Initiating a Subscription

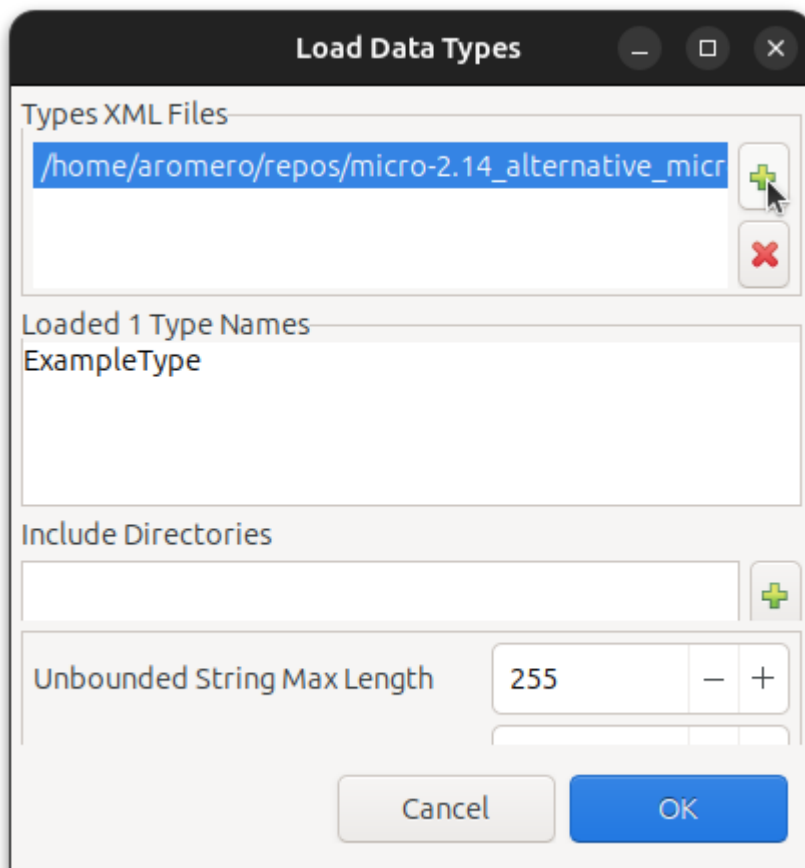
- In the "DDS Logical View," right-click on the desired topic.



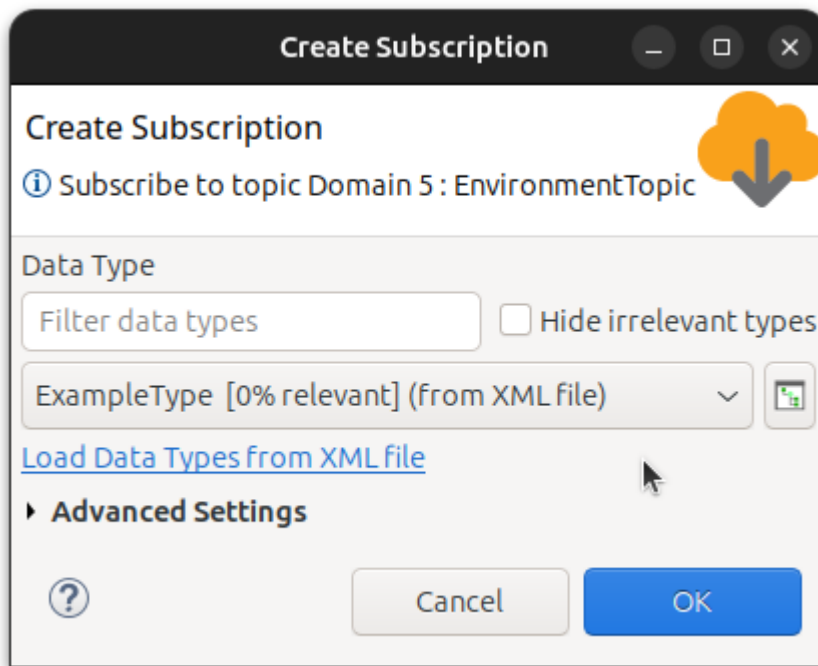
- Select "Subscribe".
- A new window will let you configure the subscriber options. Since Micro doesn't propagate the Type definitions, you need to load them from the DDS-XML file corresponding to the Micro application currently in use. Select "Load Data Types from XML file", click the "+" symbol, and select the XML file



that contains the datatypes.

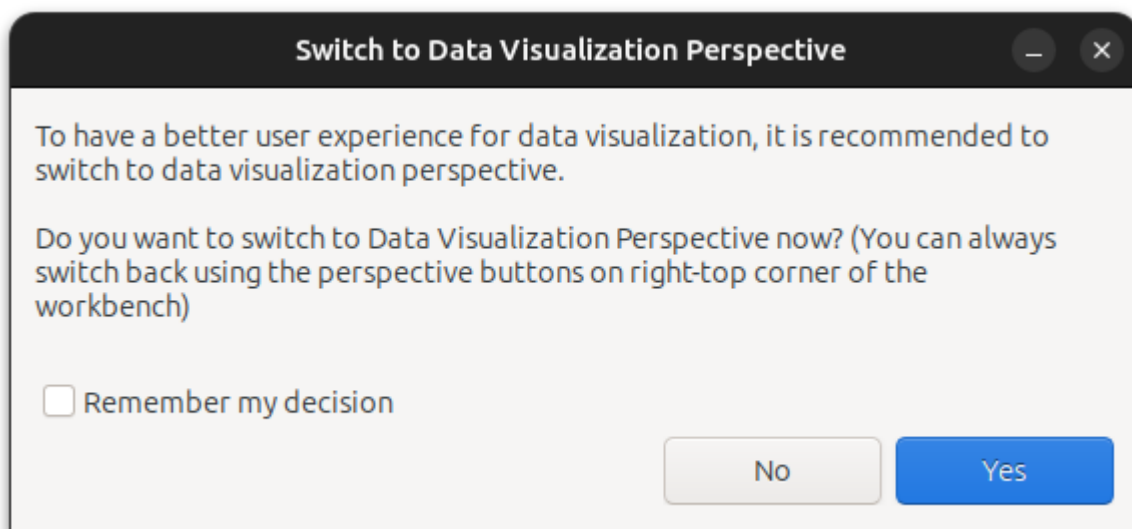


4. After adding the file, **uncheck "Hide irrelevant types"** and you should be able to find the type in the dropdown menu. Click "OK" to start subscribing to data.

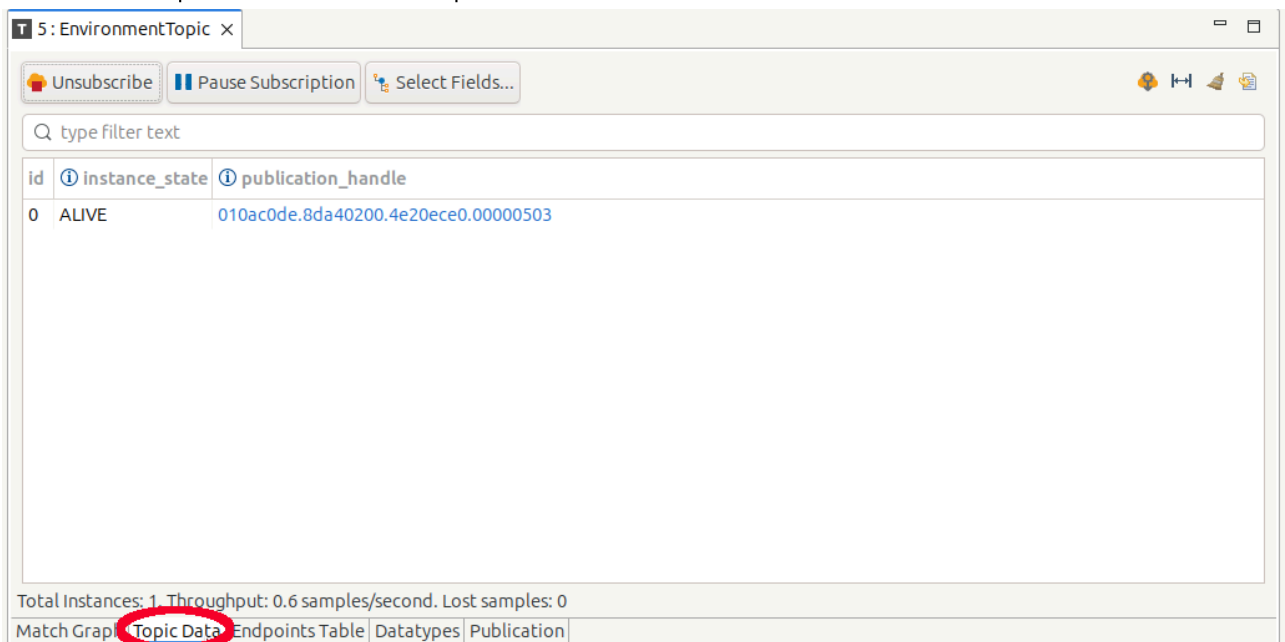


2.2 Confirming reception of data (Validating Data Published by the MCU)

1. Admin Console will ask you to switch to the "Visualization Perspective", click yes. (If you later want to go back to the "Administration Perspective", click Administration button  on the top-right corner)



2. Select the “Topic Data” tab in the Topic view. You will find a table with the last value of each instance.



5: EnvironmentTopic

Unsubscribe Pause Subscription Select Fields...

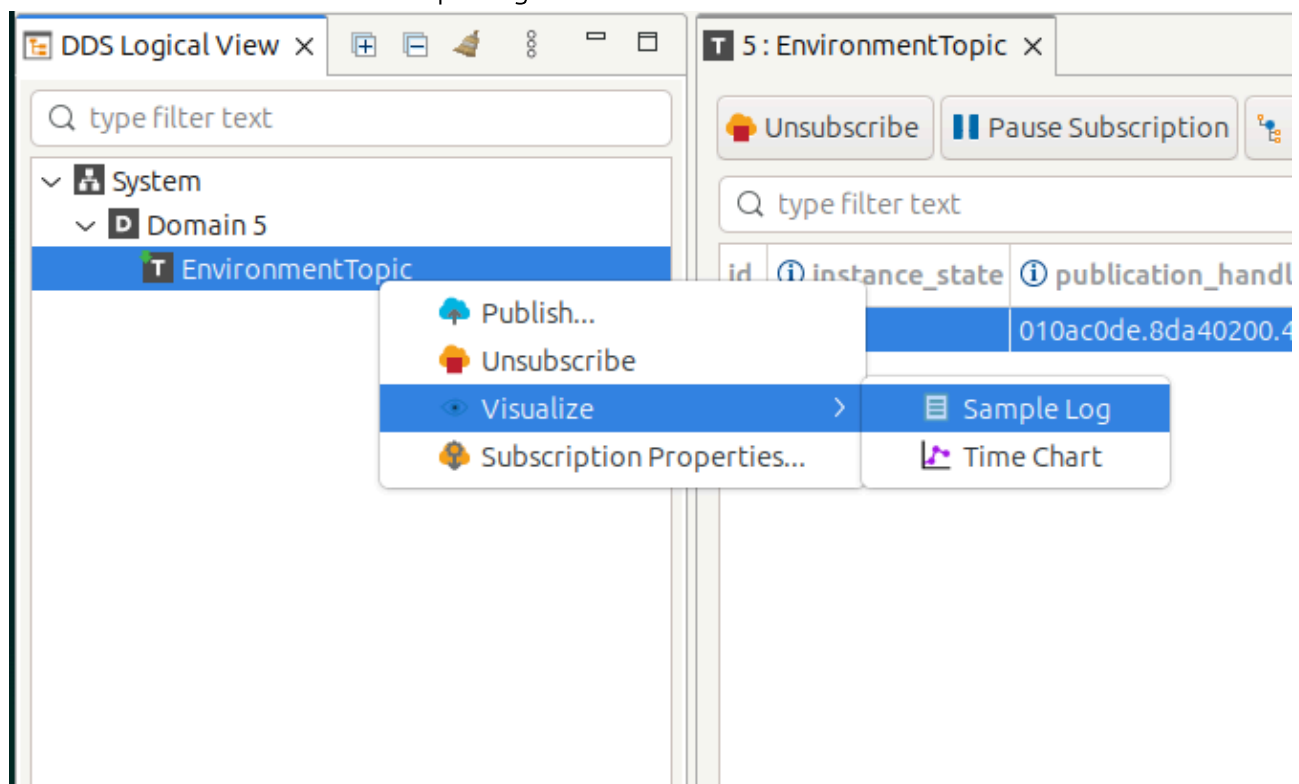
Q type filter text

id	instance_state	publication_handle
0	ALIVE	010ac0de.8da40200.4e20ece0.00000503

Total Instances: 1. Throughput: 0.6 samples/second. Lost samples: 0

Match Graph **Topic Data** Endpoints Table Datatypes Publication

3. To confirm that samples are being constantly published, right click on the topic from the “DDS Logical View” and select “Visualize”>“Sample Log”



DDS Logical View

Q type filter text

- System
 - Domain 5
 - EnvironmentTopic**

5: EnvironmentTopic

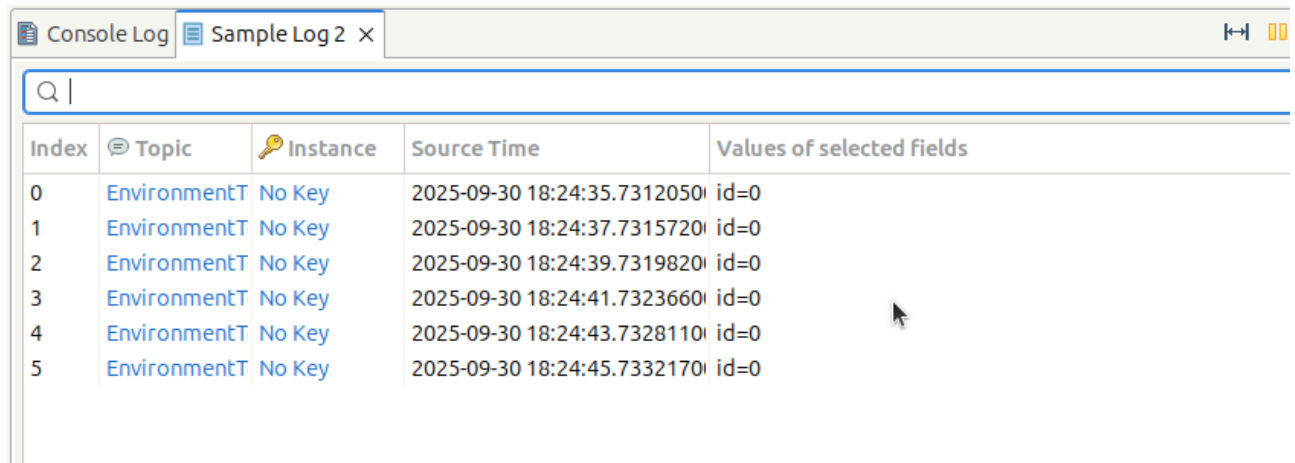
Unsubscribe Pause Subscription

Q type filter text

id	instance_state	publication_handle
		010ac0de.8da40200.4

- Publish...
- Unsubscribe
- Visualize** >
 - Sample Log
 - Time Chart
- Subscription Properties...

4. The Sample Log shows each sample received since it was opened, **if you see samples being added, the Micro application is currently publishing, therefore validation is OK.**



Index	Topic	Instance	Source Time	Values of selected fields
0	EnvironmentT	No Key	2025-09-30 18:24:35.7312050	id=0
1	EnvironmentT	No Key	2025-09-30 18:24:37.7315720	id=0
2	EnvironmentT	No Key	2025-09-30 18:24:39.7319820	id=0
3	EnvironmentT	No Key	2025-09-30 18:24:41.7323660	id=0
4	EnvironmentT	No Key	2025-09-30 18:24:43.7328110	id=0
5	EnvironmentT	No Key	2025-09-30 18:24:45.7332170	id=0

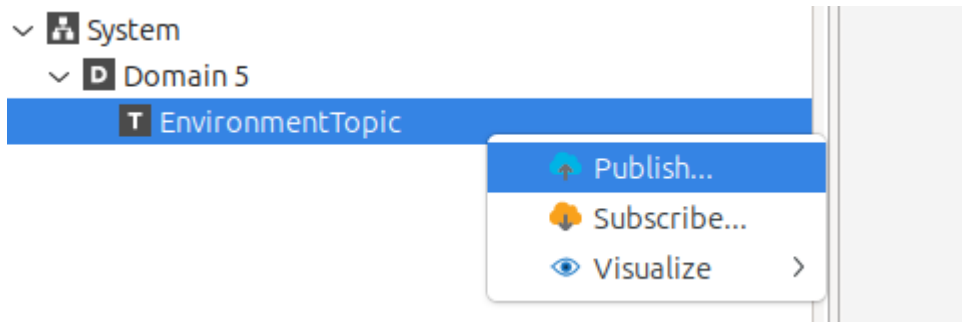
You can click any of them to get its content and metadata

3. Publishing Data Using Admin Console

The Admin Console can also be used to publish data to topics, which is useful for validating data reception in your Connex Micro application.

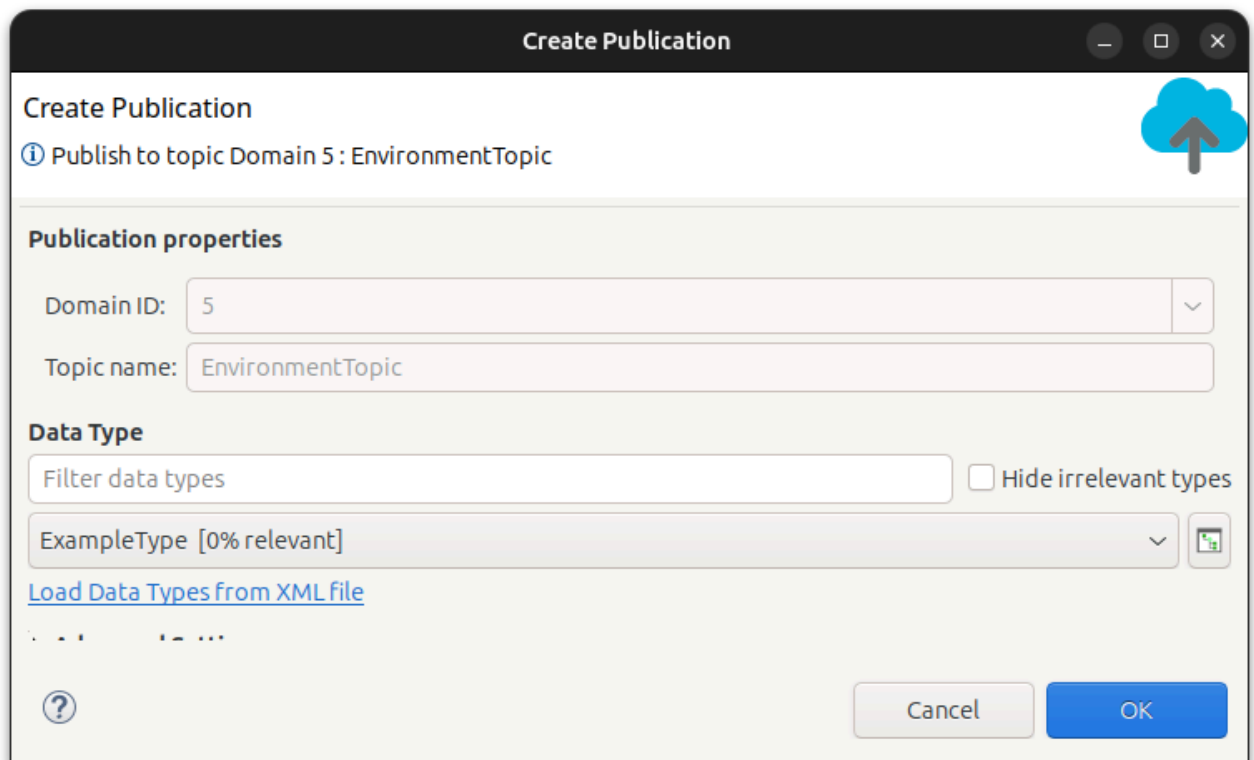
3.1 Creating a Publisher

1. In the "DDS Logical View," right-click on the desired topic.



2. Select "Publish".
3. A new window will let you configure the publisher options similar to the one for subscribing. Since Micro doesn't propagate the Type definitions, you need to load them from the DDS-XML file corresponding to the Micro application currently in use. Select "Load Data Types from XML file", click the "+" symbol, and select the XML file that contains the datatypes. Set the correct Data Type for the

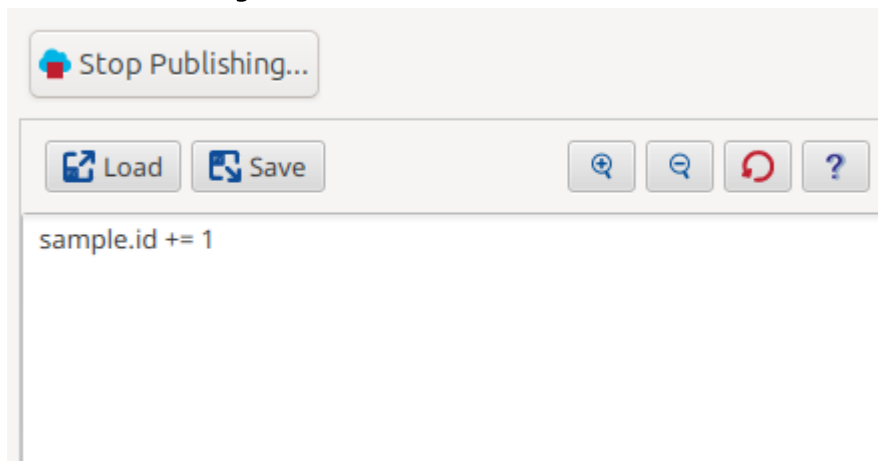
topic.



The "Create Publication" dialog box is shown. It has a title bar with standard window controls. The main content area is titled "Create Publication" and includes a sub-header "Publish to topic Domain 5 : EnvironmentTopic" with a cloud icon. Below this, the "Publication properties" section contains two input fields: "Domain ID" with the value "5" and a dropdown arrow, and "Topic name" with the value "EnvironmentTopic". The "Data Type" section features a "Filter data types" input field, a "Hide irrelevant types" checkbox, and a dropdown menu showing "ExampleType [0% relevant]" with a small icon. A link "Load Data Types from XML file" is present below the dropdown. At the bottom, there is a question mark icon, a "Cancel" button, and an "OK" button.

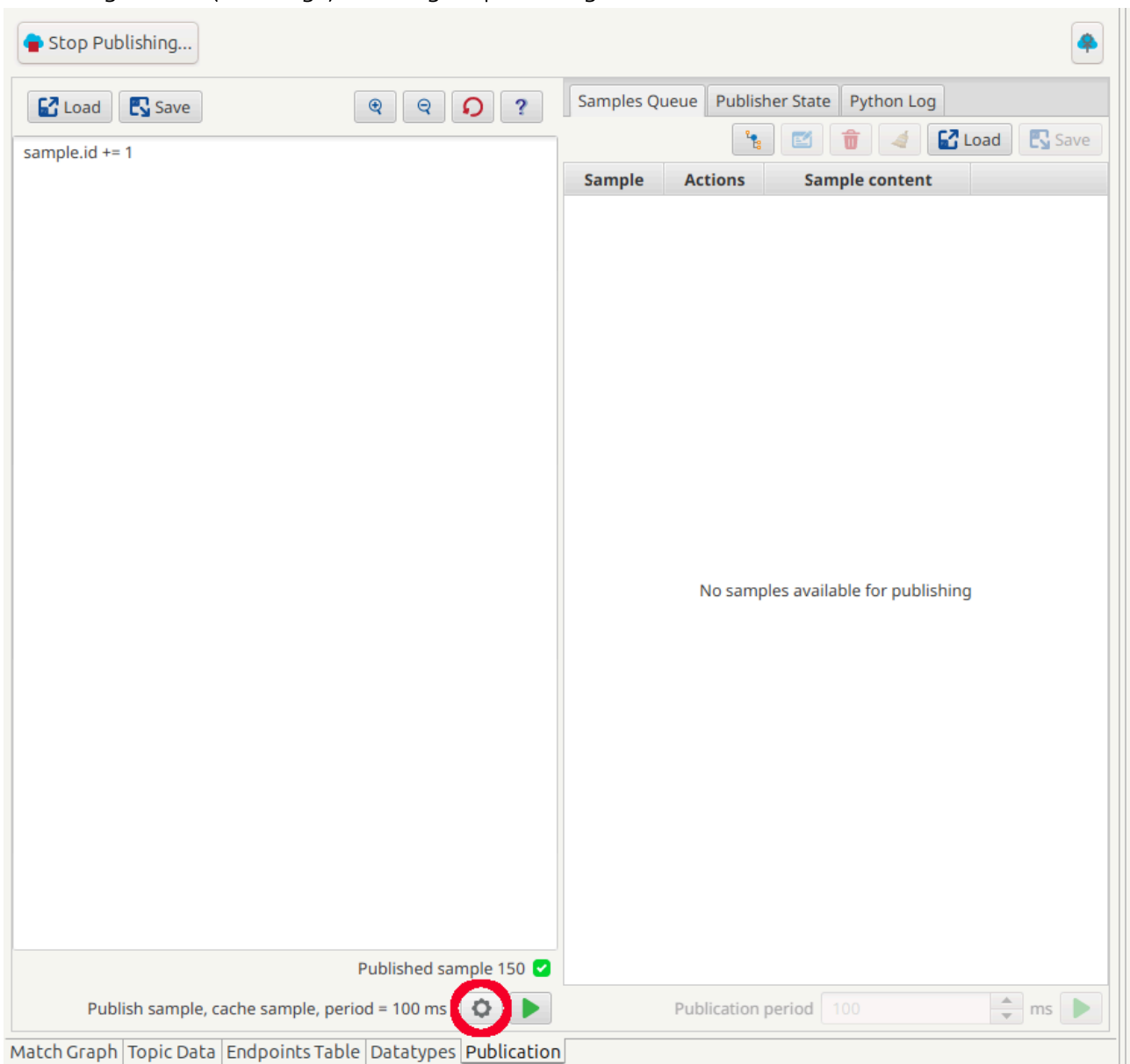
3.2 Publishing

1. In the "Publication" tab of the topic, you will see a template for the topic's data type.
2. Fill in the fields with the desired data values.
3. The publication module runs a python script. To validate the reception of data in the MCU. Make some of the values change in each iteration:

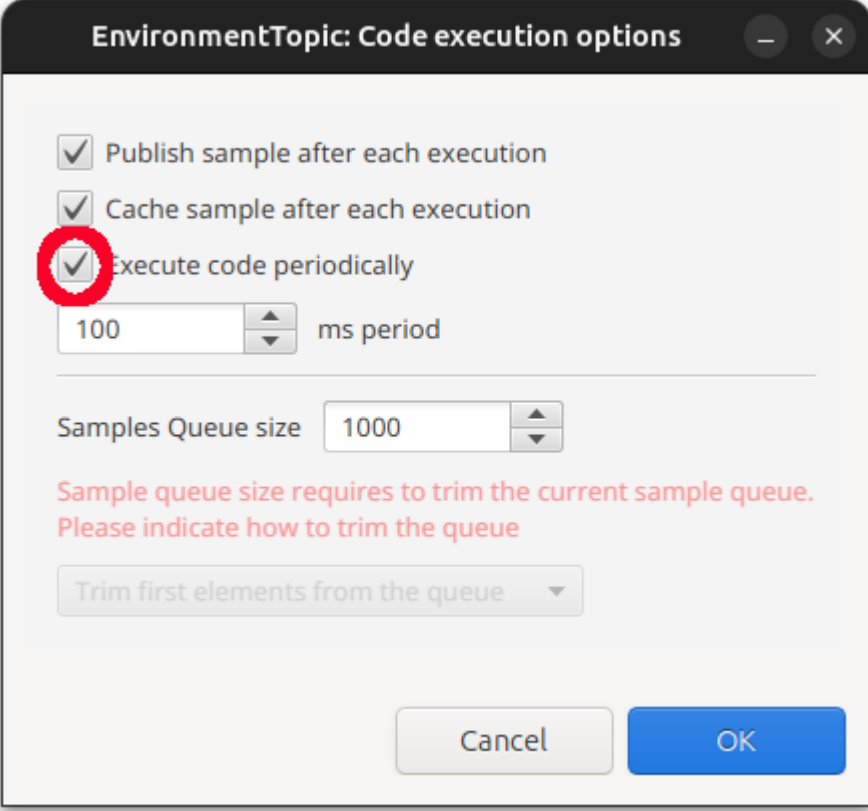


The "Stop Publishing..." dialog box is shown. It has a title bar with a cloud icon and the text "Stop Publishing...". Below the title bar, there are two buttons: "Load" and "Save". To the right of these buttons are four icons: a magnifying glass, a magnifying glass with a minus sign, a circular arrow, and a question mark. Below the buttons and icons, the text "sample.id += 1" is displayed.

4. Click the gear icon (see image) to configure publishing



5. In the Code Execution Options, check the “Execute code periodically” option.



EnvironmentTopic: Code execution options

☒ Publish sample after each execution

☒ Cache sample after each execution

☒ Execute code periodically

100 ms period

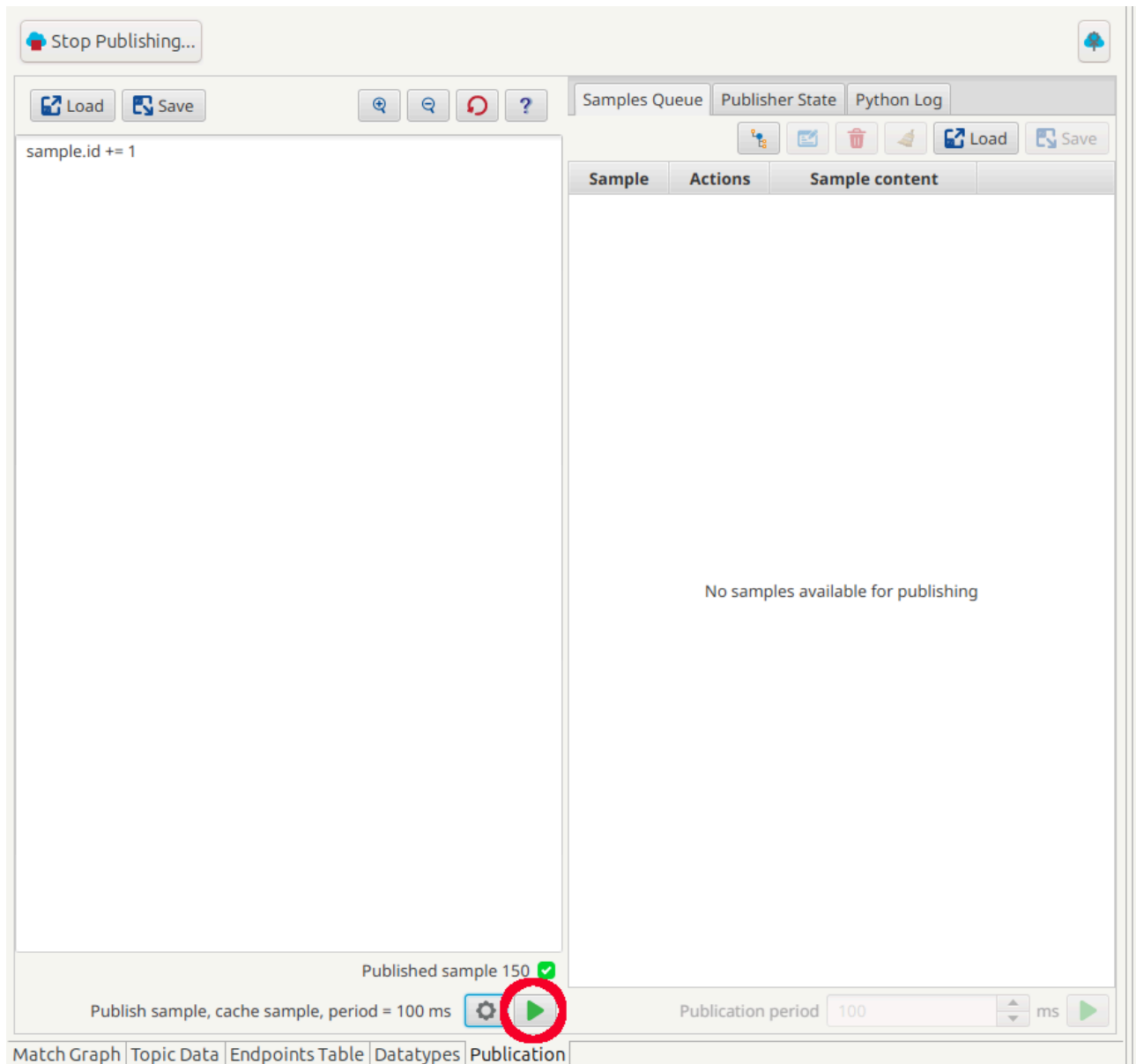
Samples Queue size 1000

Sample queue size requires to trim the current sample queue.
Please indicate how to trim the queue

Trim first elements from the queue

Cancel OK

6. Run the code by clicking the Play icon. Admin Console will start publishing samples using the configured period.



3.3 Validate reception of data in the MCU

1. If the device has an output, and you have access to it. Confirm that the string printed after receiving correct samples is printed. **If the string is outputted, validation is OK**
2. If you don't have access to an output, use a debugger to watch the content of the sample variable. Since Admin Console is publishing different values each time, **if you see the values changing, validation is OK.**

Annex {#annex}

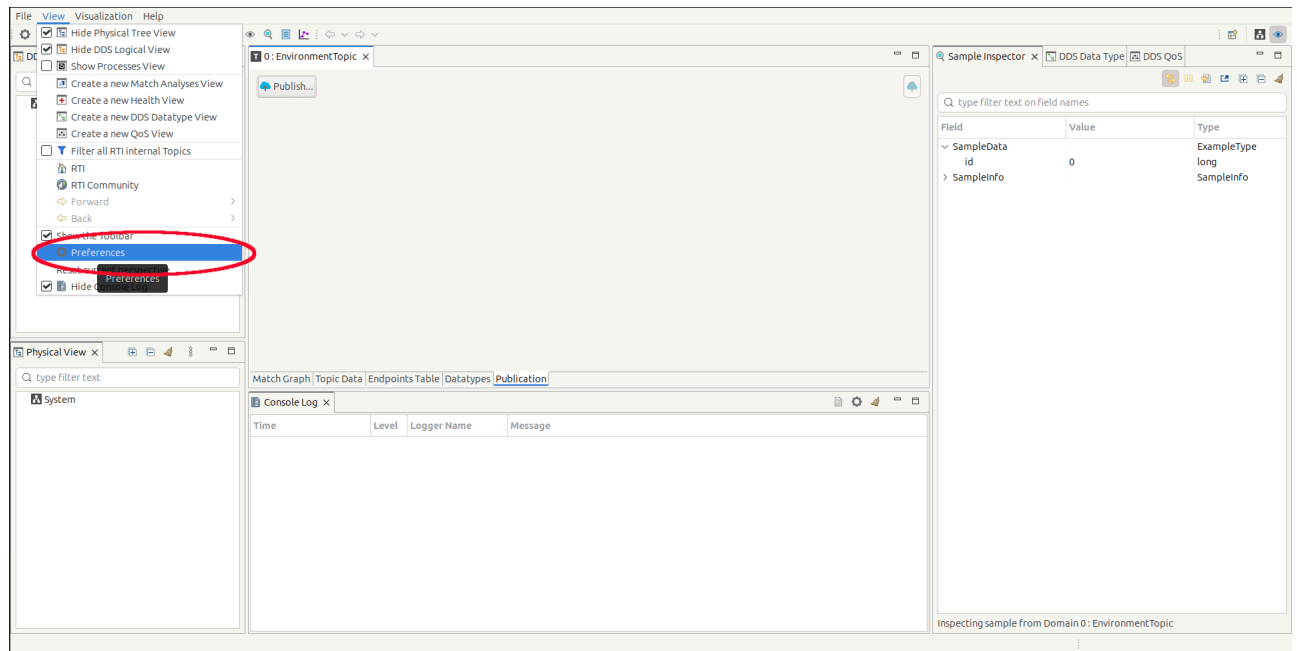
Configuring Admin Console QoS

The previous guide explains how to validate an example application using a simple DDS-XML. To use a production DDS-XML, you may need to configure some QoS in Admin console. This is because some datatypes may make discovery packets too large for being received in some of the MCUs.

Along with this guide, you will find an XML configuration file called *Ac qos.xml* (shipped at the same level as this document). This file contains the configuration needed to avoid the mentioned problem with MCUs

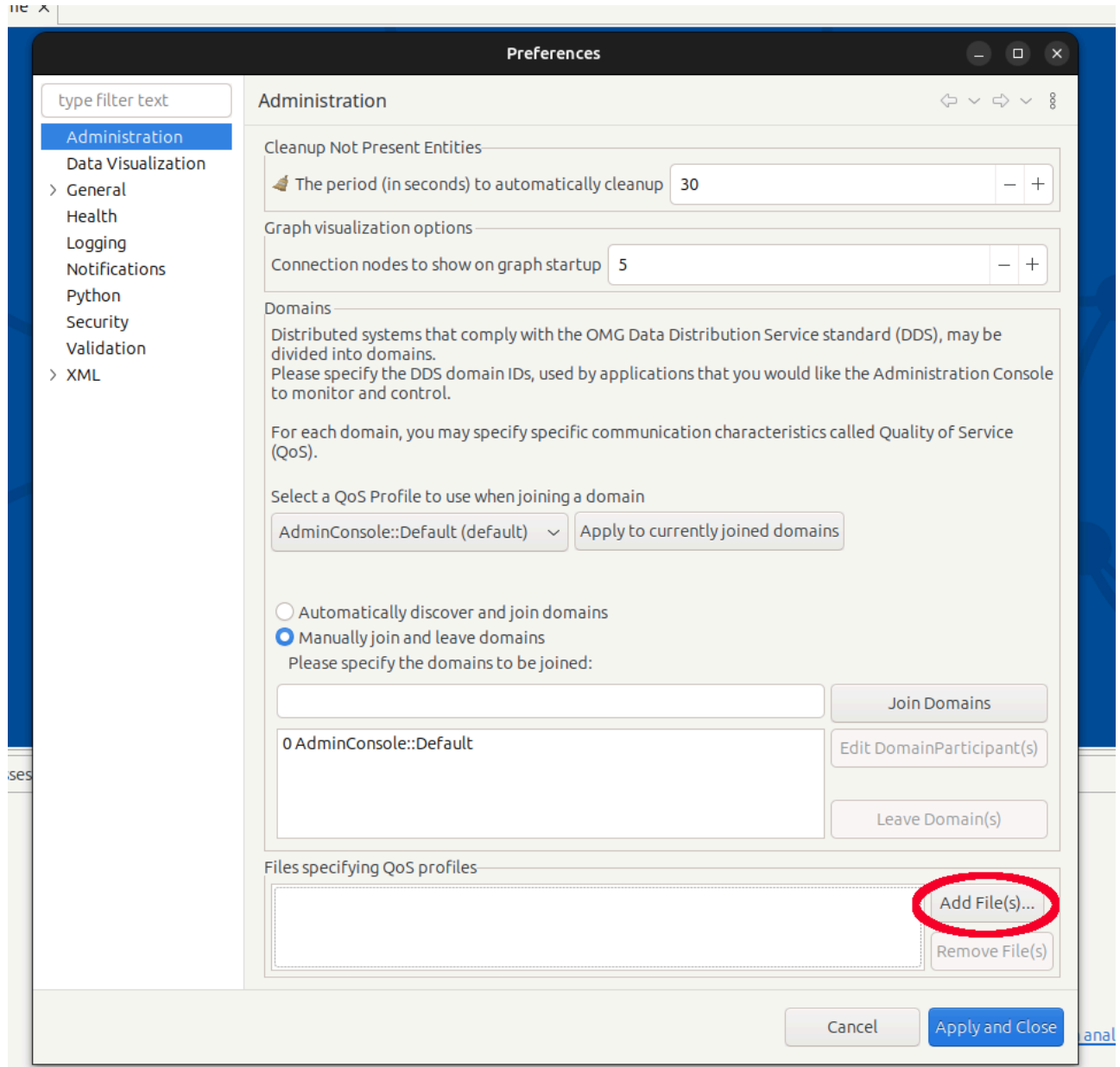
using custom DDS-XML. Follow these instructions to use it in Admin Console.

1. Go to View > Preferences.

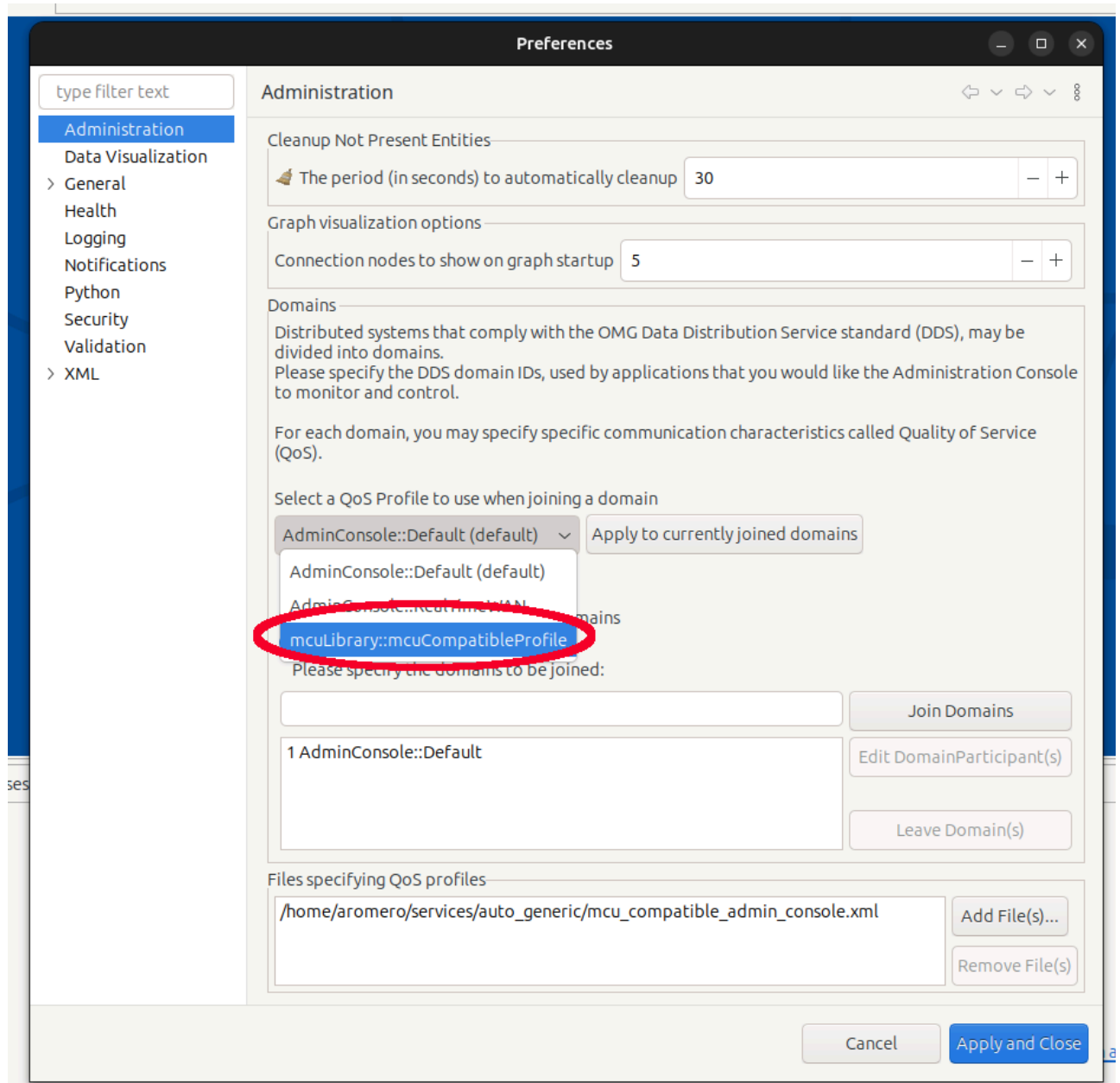


2. In the Preferences dialog, select the Administration tab.

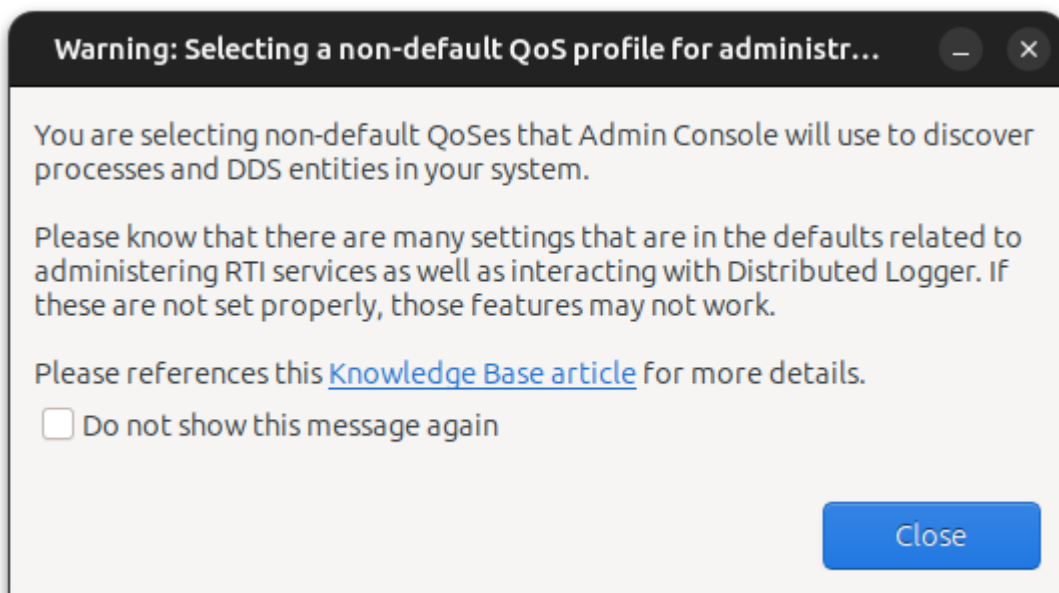
3. In the bottom part of the window, click on "Add File(s)...". Select the provided *Ac qos.xml* file.



4. Select "mcuLibrary::mcuCompatibleProfile" as the profile to use when joining a domain.



5. A warning will appear, this is expected, close it.



6. Click on "Apply to currently joined domains"

The screenshot shows the 'Preferences' dialog box with the 'Administration' tab selected. The 'Domains' section is expanded, showing options for domain management. The 'Apply to currently joined domains' button is circled in red.

Administration

Cleanup Not Present Entities

The period (in seconds) to automatically cleanup 30

Graph visualization options

Connection nodes to show on graph startup 5

Domains

Distributed systems that comply with the OMG Data Distribution Service standard (DDS), may be divided into domains. Please specify the DDS domain IDs, used by applications that you would like the Administration Console to monitor and control.

For each domain, you may specify specific communication characteristics called Quality of Service (QoS).

Select a QoS Profile to use when joining a domain

mcuLibrary::mcuCompatibleProf **Apply to currently joined domains**

☐ Automatically discover and join domains

☒ Manually join and leave domains

Please specify the domains to be joined:

1 AdminConsole::Default

Join Domains

Edit DomainParticipant(s)

Leave Domain(s)

Files specifying QoS profiles

/home/aromero/services/auto_generic/mcu_compatible_admin_console.xml

Add File(s)...

Remove File(s)

Cancel Apply and Close

7. When joining new domains, make sure the profile is shown next to the domain ID:

The screenshot shows the 'Domains' section in the Preferences dialog box. The list of domains is updated to show the profile next to the domain ID.

2 mcuLibrary::mcuCompatibleProfile

1 mcuLibrary::mcuCompatibleProfile

Join Domains

Edit DomainParticipant(s)

Leave Domain(s)